VMware L3 Interview Preparation Guide

vCenter | ESXi | vMotion | HA | DRS | Storage | Networking | Horizon VDI

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L3 Interview Success Tips

• Always mention the specific log file you would check first. • Use structured answers: Identify → Isolate → Resolve → Verify. • Mention impact and business continuity considerations. • Reference real examples from your HCLTech/TCS experience.

Topic: vCenter

Q1	vCenter Server service is not starting. How do you troubleshoot?
▶	Check vCenter services: VMware vCenter Server, VMware vSphere Profile-Driven Storage, VMware VPXD, VMware VIM Management Agents.
	Review logs: /var/log/vmware/vpxd/vpxd.log for errors or DB connection failures.
	Verify the embedded PostgreSQL or external DB (MS SQL/Oracle) is running and reachable.
	Check disk space on the vCenter appliance — full disk is a common culprit.
	Use the vCenter VAMI (https://<vcenter>:5480) to restart services and check appliance health.
	If VCSA, run: service-control --status --all and service-control --start --all.
	Validate SSL certificates — expired certs can block vCenter from starting.
	Check DNS resolution: vCenter must resolve its own FQDN and the PSC FQDN correctly.
💡	**Pro Tip:** *Always check vpxd.log first — it contains the most specific error about why vCenter failed to start.*

Q2	vCenter loses connectivity to all ESXi hosts simultaneously. What do you check?
▶	Check if vpxa (vCenter agent on ESXi) is running on the hosts: /etc/init.d/vpxa status.
	Verify management network on vCenter: ping ESXi hosts from vCenter.
	Check DNS — if vCenter cannot resolve ESXi hostnames, all hosts will show as disconnected.
	Review vCenter's own management NIC and default gateway settings.
	Look at vpxa.log on an ESXi host (/var/log/vpxa.log) for reconnect errors.
	Restart vCenter services: service-control --restart --all.
	Check for certificate expiry — expired Machine SSL cert will cause mass disconnection.
	If time drift >5 minutes between vCenter and ESXi hosts, Kerberos auth fails — verify NTP.
💡	**Pro Tip:** *NTP sync issues and expired SSL certificates are the top two causes of mass host disconnection from vCenter.*

Q3	How do you recover vCenter from a corrupted or failed state?
▶	If VCSA: use vCenter File-Based Backup (FBB) to restore from the last backup via VAMI.
	If using vSphere Replication or third-party backup (Veeam), restore the VCSA VM to a snapshot.
	In a pinch, redeploy VCSA from ISO and perform a Restore during Stage 2 setup by pointing to the backup.
	After restore, verify VC services are all running: service-control --status --all.
	Reconnect ESXi hosts that dropped from inventory and ensure datastores are visible.
	Re-register VMs from datastores if inventory was lost.
	Validate SSO, PSC, and inventory data integrity post-restore.
💡	**Pro Tip:** *Always maintain a regular vCenter backup schedule using FBB or Veeam. Target RPO should be daily or better for production vCenters.*

Q4	What is the vCenter Enhanced Linked Mode and how do you troubleshoot replication issues?
▶	Enhanced Linked Mode joins multiple vCenter Servers through a common SSO domain, allowing unified inventory view.
	Replication happens via VCHA (vCenter HA) or through PSC replication in older deployments.
	If inventory is out of sync, check the vmdir (VMware Directory Service) replication: /usr/lib/vmware-vmdir/bin/vdcrepadmin.
	Review /var/log/vmware/vmdir/vdcrepadmin.log for replication lag or errors.
	Ensure all vCenter nodes can communicate over port 2012 (LDAP), 2020 (replication).
	Use 'vdcrepadmin -f showpartners' to list replication partners and their status.
	Restart the vmdir service if replication is stuck: service vmdir restart.
💡	**Pro Tip:** *Use 'vdcrepadmin -f showstatus' to check replication agreement status between PSC nodes before escalating.*

Topic: ESXi Host

Q5	VMware VM is powered off and cannot start. What should be checked?
▶	Verify datastore has sufficient free space — ESXi needs space for swap files when powering on VMs.
	Confirm VM files are intact: VMX (config), VMDK (disk), NVRAM (BIOS state) — check for missing or locked files.
	Review ESXi host resource availability: CPU ready time, memory overcommit ratio.
	Check /var/log/vmkernel.log for errors like 'Cannot allocate memory' or disk path failures.
	Look at /var/log/vmware.log inside the VM's folder for VM-specific power-on failures.
	Verify that no snapshot consolidation is pending — this can block power-on.

	If VM is on shared storage (SAN/NAS), confirm LUN/NFS mount is accessible from the host.
	Check if another VM process has locked the VMDK using: vmkfstools -D <vmdk path>.
💡	**Pro Tip:** *vmware.log inside the VM's directory has the most specific error. Always check it alongside vmkernel.log.*

Q6	ESXi host reports a Purple Screen of Death (PSOD). What are your next steps?
▶	Do not power off immediately — capture a photo of the full PSOD screen (includes backtrace and fault address).
	After reboot, collect the core dump from /var/core/ and the ESXi logs from /var/log/.
	Review vmkernel.log for errors in the minutes leading up to the crash.
	Check hardware health: IPMI/iLO/iDRAC for CPU/RAM failures, disk errors, or thermal events.
	Validate ESXi version compatibility with host hardware via VMware HCL (Hardware Compatibility List).
	Check for driver/firmware mismatch — especially NIC, HBA, and storage controller drivers.
	Apply latest ESXi patches and firmware updates after identifying root cause.
	Open a VMware Support Request and attach the vm-support bundle: vm-support -z.
💡	**Pro Tip:** *The PSOD backtrace message (e.g., 'NMI IPI', 'LPD', 'assert') is the key to root cause — Google the exact string or raise with VMware GSS.*

Q7	ESXi host shows 'Disconnected' in vCenter. How do you recover it?
▶	Ping the ESXi management IP from the vCenter — confirm basic network connectivity.
	SSH to the ESXi host and check management agents: /etc/init.d/hostd status, /etc/init.d/vpxa status.
	Restart management agents: /etc/init.d/hostd restart && /etc/init.d/vpxa restart.
	Or use DCUI (F2 menu) > Troubleshooting Options > Restart Management Agents.
	Check DNS: ESXi must resolve vCenter FQDN and vice versa.
	If agents restart but host remains disconnected, try removing and re-adding the host to vCenter.
	Review /var/log/vpxa.log for connection refused or SSL errors.
	Check for time drift between ESXi and vCenter — must be within 5 minutes for Kerberos auth.
💡	**Pro Tip:** *Always restart hostd before vpxa — hostd must be running for vpxa to connect successfully.*

Q8	ESXi host has high CPU usage but VMs are idle. What is happening?
▶	Check for CPU overcommitment: too many vCPUs allocated vs available physical CPU cores.
	Review '%RDY' (CPU Ready) in vSphere performance charts — above 5% indicates contention.
	Check if hyper-threading is enabled: oversubscription with HT can cause cache thrashing.
	Use esxtop (press 'c' for CPU view) — look for CSTP% and RDY% at VM level.
	Identify VMs with high vCPU:pCPU ratio and right-size them.

	Look for stuck or zombie VM tasks consuming CPU cycles: esxcli vm process list.
	Check if VMware Tools is outdated — poor balloon driver performance can mask itself as CPU load.
	Consider using DRS to rebalance VMs across cluster hosts.
💡	**Pro Tip:** *In esxtop, CSTP% (co-stop) above 3% indicates SMP VMs are waiting for co-scheduling — reduce vCPU count on those VMs.*

Q9	'File Locked' errors occurring on VM VMDK. How do you identify and release the lock?
▶	Run: vmkfstools -D <path-to-vmdk> to identify which host/MAC holds the lock.
	The output shows the MAC address of the ESXi host holding the lock — map it to find the host.
	If the locking host has crashed, the lock remains stale — reboot that host to release it.
	If the VM is running on another host (split-brain), power off the VM safely and consolidate.
	For NFS datastores, the lock owner is visible in the .lck file next to the VMDK.
	Never delete the .lck file manually on VMFS — it can corrupt the VMDK.
	After releasing the lock, power on the VM and verify disk consistency.
💡	**Pro Tip:** *The MAC address from vmkfstools -D maps to the locking ESXi host's management NIC MAC — use 'esxcli network nic list' to cross-check.*

Topic: vMotion

Q10	vMotion fails with 'A general system error occurred'. How do you troubleshoot?
▶	Check vMotion network: ensure VMkernel adapter tagged for vMotion exists on both source and destination hosts.
	Verify vMotion network has L2 adjacency between hosts (same VLAN or valid routing for vMotion traffic).
	Review /var/log/vmkernel.log on both hosts for the exact migration error.
	Confirm no resource constraints on destination: sufficient CPU, RAM, and datastore space.
	Check EVC mode — if hosts have different CPU generations without EVC, vMotion will fail.
	Validate that VM's network port group exists on the destination host.
	Test vMotion bandwidth using vmkping: vmkping -I vmk1 -d -s 1472 <dest-host-vmotion-ip>.
	Ensure firewall rules allow vMotion traffic (port 8000) between hosts.
💡	**Pro Tip:** *Always validate both the vMotion VMkernel adapter AND the port group VLAN on the destination host before migration.*

Q11	VM network connectivity fails after vMotion. What should be checked?
▶	Verify the VM is connected to the correct port group on the destination host.

	Check VLAN ID consistency: source and destination port groups must have identical VLAN tags.
	Ensure physical NIC uplinks on the destination host are active and connected to the same switches.
	Check if Notify Switches is enabled on vSwitch — helps update physical switch MAC tables post-migration.
	Verify DVS (Distributed vSwitch) configuration: all hosts must use the same DVS version and port group settings.
	On the guest OS, try releasing and renewing IP (ipconfig /release && ipconfig /renew or dhclient).
	Check if CDP/LLDP shows the correct physical switch port is carrying the right VLAN.
	If using NIC teaming, verify load balancing policy is consistent across hosts.
💡	**Pro Tip:** *Enable 'Notify Switches' on vSwitch settings — it sends RARP packets to update CAM tables on physical switches after vMotion.*

Q12	What is the difference between vMotion, Storage vMotion, and Cross-vCenter vMotion?
▶	vMotion: Migrates a running VM's compute (memory + CPU state) from one ESXi host to another with zero downtime. Storage remains in place.
	Storage vMotion (svMotion): Migrates a running VM's disk files (VMDKs) from one datastore to another. VM can stay on the same host. Zero downtime.
	Cross-vCenter vMotion: Migrates a VM (compute + optionally storage) between two different vCenter Servers. Requires vSphere 6.0+ and Enhanced Linked Mode or a common SSO domain.
	Long-Distance vMotion (vSphere 6.0+): Allows vMotion across different L3 networks and even geographies using stretched clusters.
	Requirements for vMotion: Shared storage (or use Storage vMotion), vMotion VMkernel adapter, compatible CPU (or EVC), identical port groups.
	Storage vMotion can run simultaneously with vMotion — called a 'full migration' (compute + storage together).
💡	**Pro Tip:** *For Cross-vCenter vMotion in vSphere 7+, vCenter Servers no longer need to be in Enhanced Linked Mode — they just need network reachability.*

Q13	vMotion takes too long or gets stuck at 14%. What are the causes?
▶	14% stuck is classic memory copy phase issue — VM has high memory activity (dirty pages generated faster than copied).
	Check vMotion network bandwidth — minimum 1Gbps recommended; 10Gbps for large or memory-intensive VMs.
	The VM may have memory reservations or large working set that slows the iterative copy phase.
	Check for high guest memory activity (IOPS, in-memory databases) — reduce load if possible before migration.
	Verify no CBRC (Content-Based Read Cache) is conflicting with Storage vMotion.

	Increase vMotion traffic priority on DVS: set Network I/O Control reservation for vMotion traffic.
	If stuck for >1 hour, the migration will time out automatically — consider scheduling during low activity window.
	Use multiple vMotion VMkernel adapters (multiNIC vMotion) for higher bandwidth.
💡	**Pro Tip:** *Multi-NIC vMotion (vSphere 5.1+) can use up to 16 VMkernel adapters simultaneously to maximize migration bandwidth.*

Topic: High Availability (HA)

Q14	vSphere HA fails to restart VMs after a host failure. What are possible reasons?
▶	Insufficient failover capacity: HA admission control policy may not allow restart if cluster lacks resources.
	Check HA admission control settings: percentage-based or slot-based policy may be too restrictive.
	VM restart priority may be set to Disabled — check VM-level HA overrides.
	Verify HA is enabled on both the cluster and the individual VMs.
	Check if VMs have memory reservations — slots are calculated based on reservation values.
	Ensure the remaining hosts have enough CPU and RAM to absorb the failed host's VMs.
	Review /var/log/fdm.log (Fault Domain Manager) on surviving hosts for HA election and restart events.
	Check if HA heartbeat datastores are accessible — HA uses datastore heartbeating for network partition detection.
💡	**Pro Tip:** *fdm.log is the key HA log — it shows master election, VM restart decisions, and failover reasons in detail.*

Q15	What is HA Admission Control and how does it work?
▶	Admission Control is an HA policy that reserves cluster resources to ensure VMs can restart after host failure.
	Three policies: (1) Cluster resource percentage — reserve X% of CPU/RAM. (2) Slot policy — reserves resources based on largest VM slot. (3) Dedicated failover hosts — specific host reserved for failover.
	If Admission Control blocks a power-on, it means the cluster cannot guarantee failover capacity.
	Slot size is calculated as: max(all VM CPU reservations) for CPU, max(all VM memory reservations + overhead) for RAM.
	A large VM with high reservation can inflate the slot size and artificially reduce cluster capacity.
	Percentage-based policy (default in vSphere 6.5+) is more flexible and avoids slot inflation.
	Admission Control can be set to 'Disabled' (not recommended for production) to allow overcommitment.

💡	**Pro Tip:** *In vSphere 6.5+, use percentage-based Admission Control with a value matching N/total hosts (e.g., 25% for N+1 in a 4-host cluster).*

Q16	How does vSphere HA detect host failure vs. network isolation?
▶	HA uses heartbeating: each host sends heartbeats to the cluster master every second via the management network.
	If a host misses heartbeats for 10 seconds (default), it is declared failed.
	But HA also uses datastore heartbeating: the host writes to a heartbeat datastore (VMFS volume). If alive but isolated, it can still heartbeat via storage.
	Host isolation response: if a host loses management network but can still heartbeat via datastore, it is 'partitioned' not 'failed' — VMs may not be restarted to avoid split-brain.
	Isolation response options: Leave powered on (default), Power off and restart, Shut down and restart.
	The master HA host monitors slave hosts and makes restart decisions.
	HA uses two heartbeat datastores per host for redundancy — configure these to be on different LUNs.
💡	**Pro Tip:** *Set Isolation Response to 'Shutdown and restart' for mission-critical VMs — 'Power off' is abrupt and skips OS shutdown.*

Q17	vSphere HA shows 'HA agent on host has an error'. How do you fix it?
▶	Review the exact error in vCenter > host > Summary > vSphere HA section.
	Common causes: management network failure, HA agent (FDM) crash, cluster misconfiguration.
	Reconfigure HA: right-click cluster > Reconfigure vSphere HA — this re-installs FDM agents.
	Check /var/log/fdm.log on the affected host for specific error messages.
	Verify management network is stable — flapping NICs cause repeated HA agent errors.
	Ensure the host can reach the master HA host via management network.
	If the host was recently added to a cluster, HA reconfiguration must complete successfully.
	Check time sync — NTP drift can cause HA cluster communication issues.
💡	**Pro Tip:** *Reconfigure HA (right-click cluster) is often the quickest fix for FDM agent errors — it's non-disruptive to running VMs.*

Topic: DRS (Distributed Resource Scheduler)

Q18	DRS is not migrating VMs even though the cluster is imbalanced. Why?
▶	Check DRS automation level: 'Manual' or 'Partially Automated' means DRS recommends but does not auto-migrate.
	Review DRS migration threshold: if set to 'Conservative' (level 1), DRS only acts on severe imbalance.

	VMs with DRS overrides set to 'Disabled' or 'Manual' will not be auto-migrated.
	Check affinity/anti-affinity rules — if rules conflict with migration, DRS will not violate them.
	Verify vMotion is configured and working — DRS requires vMotion to migrate VMs.
	VMs without VMware Tools may not be migrated by DRS in some scenarios.
	Check if the VM has memory reservations that cannot be satisfied on available hosts.
	If the cluster has only 2 hosts, DRS has limited options — imbalance may be unavoidable.
💡	**Pro Tip:** *Set DRS threshold to level 3 (Aggressive default) for balanced clusters. Level 5 migrates even for minor imbalance, which can cause unnecessary migrations.*

Q19	How does DRS calculate cluster imbalance and decide when to migrate VMs?
▶	DRS calculates a 'standard deviation' of normalized resource utilization (CPU + memory) across all hosts.
	The imbalance metric is compared against the configured DRS migration threshold.
	DRS runs every 5 minutes by default and evaluates whether migrating VMs would improve cluster balance.
	Each potential migration is given a priority (1–5) based on the benefit it provides.
	DRS also runs at VM power-on (Initial Placement) to pick the best host.
	DRS calculates a 'DRS Score' per VM showing satisfaction of resource needs.
	In vSphere 7+, per-VM DRS was introduced — DRS evaluates and acts at the individual VM level rather than cluster-wide.
	DRS respects affinity rules, anti-affinity rules, VM-host rules, and DRS overrides during decision-making.
💡	**Pro Tip:** *In vSphere 7, per-VM DRS significantly improves responsiveness — it reacts to individual VM resource changes in near real-time rather than batch every 5 minutes.*

Q20	DRS is generating too many vMotion migrations. How do you reduce unnecessary migrations?
▶	Reduce DRS aggressiveness: move threshold from level 5 toward level 1–2 (Conservative).
	Check if VM swap activity is triggering constant rebalancing — investigate memory overcommitment.
	Create VM-host affinity rules to pin specific VMs to host groups and reduce migration scope.
	Review if Proactive HA is triggering unnecessary pre-emptive migrations.
	Disable per-VM DRS if individual VM-level rebalancing is causing churn (vSphere 7+).
	Set DRS overrides for stable VMs (databases, domain controllers) to 'Partially Automated'.
	Check for host maintenance events triggering evacuation vMotions.
	Review Network I/O Control — if network is constrained, reduce vMotion traffic priority.
💡	**Pro Tip:** *For latency-sensitive VMs like databases, set DRS override to 'Partially Automated' — you approve before any live migration occurs.*

Q21	What are VM-Host Affinity Rules in DRS and when would you use them?
▶	VM-Host Affinity Rules define relationships between groups of VMs and groups of hosts.
	Types: (1) Must run on — VM group MUST run on host group (hard rule). (2) Should run on — preferred but not mandatory (soft rule). (3) Must not run on / Should not run on — anti-affinity equivalents.
	Use case: License compliance — Windows Server or Oracle licenses may require VMs to run on specific hosts.
	Use case: Hardware affinity — GPU-enabled VMs must run only on hosts with GPUs.
	Use case: Security isolation — separate sensitive VMs onto dedicated hardened hosts.
	Hard rules override DRS balancing — use with caution as they can cause cluster imbalance.
	VM-VM affinity rules: keep VMs together (co-locate) or apart (anti-affinity) — useful for app tiers or HA pairs.
💡	**Pro Tip:** *Use SOFT rules whenever possible — hard rules can leave hosts underutilized and prevent HA failover if no compliant host is available.*

Topic: VMware Storage

Q22	Datastore shows 'All Paths Down' (APD) or 'Permanent Device Loss' (PDL). How do you handle it?
▶	APD: All storage paths are unavailable but it's unclear if the loss is permanent. ESXi waits for paths to recover.
	PDL: Storage array has confirmed the device is permanently gone (SCSI SENSE code received).
	For APD: Check SAN fabric — HBA ports, zoning, switch connections. Rescan storage adapters after fixing.
	For PDL: VMs on the affected datastore will be terminated — HA can restart them if configured for PDL response.
	Configure HA to respond to PDL: set 'Response to Datastore with PDL' to 'Power Off and Restart VMs'.
	Configure HA APD response: 'Response to Datastore with APD' with timeout (default 3 minutes).
	Use SIOC (Storage I/O Control) to prioritize I/O during degraded storage conditions.
	After path recovery, verify VMDK integrity with vmkfstools -e <vmdk>.
💡	**Pro Tip:** *Enable HA PDL/APD response settings proactively — without them, VM restarts during storage failures require manual intervention.*

Q23	VMware snapshot deletion is taking too long or is stuck. How do you resolve it?
▶	Large snapshots (multiple GBs or months old) take longer to consolidate — check datastore free space first.
	Ensure datastore has at least as much free space as the largest snapshot delta file.

	Use 'Snapshot Consolidation' in vCenter (right-click VM > Snapshots > Consolidate) to merge stuck deltas.
	Check vmware.log in the VM directory for snapshot consolidation errors.
	Monitor consolidation progress in vCenter Tasks — look for 'Remove snapshot' or 'Consolidate' tasks.
	If consolidation is stuck, check if the VMDK is locked by another process: vmkfstools -D <vmdk>.
	As a last resort: power off the VM, delete snapshots, power on — consolidation is faster when VM is off.
	After consolidation, verify the snapshot chain is clean: no delta VMDKs should remain.
💡	**Pro Tip:** *Snapshots are NOT backups. Enforce a policy: no snapshot older than 72 hours in production. Use Veeam/Backup solution for actual backups.*

Q24	VM shows 'Inaccessible' after a storage migration. How do you troubleshoot?
▶	Verify the datastore is mounted and accessible on the host: esxcli storage filesystem list.
	Check if the VMX file path changed during migration — vCenter inventory may point to old path.
	Browse the datastore in vCenter to confirm all VM files (VMX, VMDKs) are present.
	Re-register the VM: remove it from inventory, then Add to Inventory by browsing to the VMX file.
	Check /var/log/vmkernel.log for I/O errors to the datastore during migration.
	If using SAN, verify LUN is visible on all hosts: esxcli storage core device list.
	Check VMFS signature — if LUN was resigned, the datastore may need to be re-mounted.
	Run VMFS volume check: vmkfstools --fix <volume path> to check for filesystem corruption.
💡	**Pro Tip:** *After Storage vMotion completes, always verify the VM powers on successfully and check disk consistency before considering the task done.*

Topic: VMware Networking

Q25	VM loses network connectivity after ESXi host NIC fails. How do you ensure resilience?
▶	Configure NIC teaming on vSwitch: use at least 2 physical uplinks (Active/Active or Active/Standby).
	Set failover detection to 'Link Status Only' or 'Beacon Probing' (more reliable, detects upstream failures).
	Beacon Probing sends probe packets from all adapters — if probes from one adapter fail, it's marked failed.
	For DVS: configure NIC teaming policy at port group level for granular control.
	Use LACP (802.3ad) for DVS if physical switches support it — provides link aggregation with failover.

	Monitor NIC status using: esxcli network nic list — check Link Status and Speed.
	Configure VMkernel adapters with redundant paths on separate physical NICs.
	Test NIC failover in maintenance window by manually disconnecting a cable and verifying traffic shifts.
💡	**Pro Tip:** *Beacon Probing detects upstream switch port failures that Link Status cannot detect — use it for critical port groups.*

Q26	What is the difference between vSphere Standard Switch (VSS) and Distributed Switch (DVS)?
▶	VSS (Standard vSwitch): Configured per-host. Changes must be repeated on each host. No centralized management. Free.
	DVS (Distributed vSwitch): Configured centrally in vCenter. Applies uniformly to all hosts in the switch. Included in vSphere Enterprise Plus.
	DVS features not in VSS: LACP, Network I/O Control (NIOC), port mirroring, IPFIX/NetFlow, per-port statistics.
	DVS supports VM-level traffic shaping; VSS only supports port group-level shaping.
	DVS stores port state in vCenter — VM network config persists across vMotion without reconfiguration.
	Migration from VSS to DVS should be done carefully — one VMkernel at a time to avoid losing management access.
	In large environments (10+ hosts), DVS is strongly preferred for operational consistency and auditability.
💡	**Pro Tip:** *Never migrate the management VMkernel adapter to DVS without a physical console backup (iLO/iDRAC) — a misconfiguration can isolate the host.*

Topic: VMware Horizon VDI

Q27	Users cannot connect to their Horizon desktops. What is your troubleshooting process?
▶	Check Horizon Connection Server status: Services > VMware Horizon View Connection Server — must be running.
	Verify the desktop pool is in an Enabled state and has available desktops.
	Check if the desktop VM is powered on and VMware Tools is running.
	Review the Horizon Administrator console for any error messages on the desktop or pool.
	Check the Composer service if using linked clones — Composer issues can prevent desktop assignment.
	Verify the PCoIP/Blast Gateway configuration — external users need correct UAG (Unified Access Gateway) routing.
	Test internal vs external connectivity — if internal works but external doesn't, issue is likely UAG/firewall.
	Review Connection Server event logs: C:\ProgramData\VMware\VDM\logs\debug*.log.

💡	**Pro Tip:** *Horizon Collector logs (debug-*.log) on the Connection Server contain session establishment errors — search for the user's username.*

Q28	Horizon linked clone desktops are showing stale data or not refreshing. How do you resolve?
▶	Linked clones use a snapshot chain from the parent VM — if the parent snapshot is outdated, desktops may lag.
	Initiate a 'Recompose' operation on the pool to update desktops with the latest parent snapshot.
	Check if the replica VM is accessible: linked clones depend on a common replica on each datastore.
	Verify View Composer is healthy: Services > VMware View Composer — check for errors.
	For persistent desktops, check if the user's profile (FSLogix or UEM) is loading correctly.
	For non-persistent desktops, verify the 'Refresh on logoff' policy is enabled in pool settings.
	Check available disk space on the datastore — linked clone growth can exhaust storage.
	Review Composer logs: C:\ProgramData\VMware\View Composer\logs\ for recompose errors.
💡	**Pro Tip:** *Always snapshot the parent VM in a clean, pre-sysprepped state before recompose — a dirty parent snapshot causes desktop provisioning failures.*

☆ Key Learnings & EVC Configuration Note

EVC (Enhanced vMotion Compatibility) — Critical Configuration Note

• EVC masks CPU features to a common baseline, enabling vMotion between hosts with different CPU generations.

• Even if vMotion works initially without EVC, CPU feature drift can cause delayed failures — especially with newer workloads using advanced CPU instructions.

• Always enable EVC at cluster creation time — changing EVC mode on a live cluster requires powering off all VMs first.

• Testing in a controlled environment before production deployment is essential to identify CPU compatibility issues early.

☑ ESXi Host Troubleshooting Quick Reference

1. Host Status	Check vCenter connectivity, CPU/memory/datastore utilization
2. Hardware Health	IPMI/iLO sensors — CPU, RAM, disk, power supply, RAID controller
3. Network Config	vSwitches, port groups, VLANs, VMkernel adapters, physical NIC status
4. Storage	Datastore availability, storage adapter rescan, multipathing, IOPS/latency
5. Review Logs	/var/log/vmkernel.log | /var/log/hostd.log | /var/log/vpxa.log

6. Mgmt Services	Restart hostd and vpxa via Shell or DCUI if host is unresponsive
7. VM-Level	Power state, VMware Tools status, snapshot usage, resource allocation
8. Patch & HCL	ESXi version current, hardware on VMware HCL, driver/firmware compatibility